

Click, Scroll, Consent: Uncovering Australia's Privacy Knowledge Crisis

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Abstract

Governments and digital platforms are intensifying their use of data collection to enable personalization, analytics, and surveillance. At the same time, ordinary citizens increasingly sense that their privacy is being violated, yet struggle to explain how or why. To investigate this issue, we present the first nationally representative study of Australians' awareness, confusion, and frustrations regarding digital privacy, consent interfaces, and the Australian Privacy Act. Drawing on a mixed-methods survey of 239 participants, we find that most Australians suspect their data is being misused but cannot identify the mechanisms behind it. More than half admit they "know something is wrong" yet cannot articulate what it is, while fewer than 20% correctly understand practices such as fingerprinting or cross-device tracking. We conceptualize this disconnect as *fractured awareness*, a socio-technical condition in which users are not only uninformed but misinformed by the very systems meant to protect them. This constitutes a critical socio-technical failure, where system design actively distorts public understanding of digital privacy practices and protections. We benchmark our findings against U.S. studies and show that Australians exhibit higher institutional trust but similar levels of functional confusion, exposing privacy myths that are culturally shaped yet globally persistent. This paper contributes to software engineering by introducing fractured awareness as a requirements-level construct for privacy-aware system design, developing privacy personas (*Misinformed Majority*, *Resigned Skeptics*, *Aware Activists*) as design archetypes, and demonstrating how regulatory opacity, dark-pattern consent mechanisms, and inequitable interfaces generate systemic confusion. We describe key implications for researchers and practitioners, including regulators, technologists, and platform designers, and argue that addressing fractured awareness is essential to restoring public trust and enabling meaningful digital consent.

CCS Concepts

• **Security and privacy** → **Privacy protections; Usability in security and privacy**; • **Software and its engineering** → **Requirements analysis**.

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Keywords

Digital Privacy, Empirical Software Engineering, Privacy-by-Design

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1 Introduction

The proliferation of digital technologies has fundamentally altered how individuals engage with platforms, services, and institutions online. Increasingly, these interactions are mediated by software that collects, tracks, analyzes, and monetizes personal data at unprecedented scales. As this intensifies, privacy has become a central concern, not only as a legal or ethical issue but as a deeply technical and socio-technical challenge that software engineers must contend with [5, 21, 22, 28, 29, 38, 39]. Yet despite efforts to encode privacy into software systems, the concept remains poorly understood by users and inconsistently implemented by developers, resulting in systems that often fail to meaningfully inform, empower, or protect individuals [8, 24, 25, 27].

This paper investigates a particularly underexplored facet of this challenge: the fractured state of public understanding we term "*fractured privacy awareness*". This refers to the condition in which individuals suspect their privacy is being violated, express frustration, anger, or resignation, yet cannot articulate the technical or legal mechanisms behind these violations. Awareness exists, but it is fragmented, characterized by emotional reactions rather than informed understanding, and intuition rather than analysis. This disconnect between suspicion and comprehension poses a significant barrier to meaningful privacy protection.

We focus on Australia, a country with comprehensive privacy legislation (the Privacy Act 1988) and ongoing reform efforts [19, 34]. Through an empirical study involving 239 diverse participants, we examine not only what Australians know about online tracking, consent mechanisms, and data usage policies, but also how they feel about their lack of clarity. Our findings reveal that while citizens strongly suspect privacy violations are occurring, they cannot explain practices such as fingerprinting or cross-device tracking [1, 16], and often describe privacy protection as a "*rigged game*" they cannot win.

Our findings suggest that these misconceptions are not merely gaps in knowledge but reflect deeper misalignments between how systems are engineered, how consent is designed, and how individuals experience privacy. For instance, over two-thirds of our respondents mistakenly believed that privacy policies prevent data sharing [25], while nearly half assumed companies must obtain

explicit opt-in consent for tracking even though such requirements are absent in most contexts [19, 34]. These insights extend critiques of “consent theatre” [26, 35], where the illusion of agency substitutes for meaningful protection.

The implications for software engineering (SE) are significant. Privacy-preserving technologies cannot succeed if users perceive them as manipulative, incomprehensible, or futile [9, 11, 23, 24, 37]. Our findings suggest that unless design practices explicitly acknowledge citizens’ fractured awareness and emotional responses, privacy by design risks remaining incomplete or even counterproductive [17]. Moreover, our comparison with U.S. data shows that Australians display a distinctive mix of higher institutional trust and equal functional confusion, underscoring the need for culturally contextualized approaches to privacy engineering [45].

By exposing the disconnect between legal frameworks, system design, and public perception, we argue for a reorientation of privacy engineering toward inclusive, comprehensible, and user-centered paradigms. Our findings highlight the urgent need for developers, policymakers, and educators to collaborate in repairing systemic misunderstandings, not only through technical safeguards or stronger regulation, but through systems that genuinely communicate and restore user agency.

This study is motivated by, and builds upon, prior work on privacy literacy, the privacy paradox, and user misconceptions [3, 10, 43, 44]. While previous studies in the United States and Europe have highlighted low comprehension, our work contributes a novel and contextually grounded perspective by:

- Conducting the first comprehensive study of privacy awareness in Australia, paralleling U.S.-based methodologies (e.g., Turow et al. [41, 42]) while also capturing the emotional landscape of suspicion, distrust, and resignation.
- Introducing the concept of fractured privacy awareness, where citizens sense violations but lack understanding, highlighting the emotional toll of systemic opacity.
- Demonstrating that Australia’s comparatively privacy-aware legal framework has fostered misplaced trust rather than genuine literacy, with many assuming protections exist where none exist.

2 Motivation

In recent years, digital privacy has evolved from a niche policy issue to a central societal concern at the intersection of technology, governance, human rights, and ethics. Software infrastructures now support nearly every aspect of daily life, including banking, health-care, education, and civic participation, while simultaneously mediating how personal data is extracted, shared, and monetized. Yet the mechanisms through which privacy is communicated, understood, and enacted in software routinely fall short of protecting individuals, particularly in democracies where formal legal frameworks exist but fail to translate into meaningful safeguards [6, 13, 36].

Australia presents a paradoxical case. It has one of the world’s longest-standing federal privacy laws, the Privacy Act 1988, and is currently undergoing an extensive reform process to modernize its provisions [32, 33]. The country has also introduced sector-specific protections such as the My Health Record Act and the Consumer Data Right. On paper, these frameworks signal a robust commitment to safeguarding citizens’ data. Legal reform continues,

but everyday Australians are left navigating a digital ecosystem marked by opacity, uncertainty, and growing resignation.

Prior studies have shown that users frequently misinterpret privacy policies [25], and conflate interface choices with legal guarantees [36]. They often acquiesce to data collection not through meaningful consent but through patterns of confusion, habituation, and what has been described as digital resignation, i.e., a sense that resistance is futile in the face of pervasive surveillance [14]. In the Australian context, the Office of the Australian Information Commissioner (OAIC) reports rising numbers of privacy complaints, particularly concerning online platforms and mobile applications [33]. Yet these complaints are rarely accompanied by concrete knowledge about how to exercise privacy rights or understand the pathways through which data circulates. This disconnect between voiced concern and practical comprehension reflects what we term “*fractured privacy awareness*”, a condition in which citizens recognize their vulnerability and suspect data exploitation, yet cannot pinpoint how, or by whom.

This disjunction between regulatory intent and public understanding creates a high-risk environment. Users are routinely expected to make decisions about consent, data disclosures, and terms of service as if they were legal or technical experts. Meanwhile, software designers and regulators continue to operate under assumptions of rational, informed users, assumptions that decades of research in HCI, law, and behavioral economics have consistently refuted [40, 46]. The resulting gap, between what users believe, what systems do, and what laws provide, is more than a usability issue. It raises fundamental questions of justice and equity, as the burdens of (mis)understanding fall disproportionately on older adults, low-income groups, and those with limited digital literacy [10, 11].

Comparative evidence highlights the urgency of this gap. In the United States, Turow et al. [42] found that most Americans held false beliefs about protections under laws such as HIPAA and COPPA, assuming strong safeguards where little existed. Similar patterns are emerging in Australia. Many people trust that federal laws prevent the sale of personal data, yet few realize that health and fitness applications can legally share or sell sensitive information to third parties [20]. Unlike Europe, Australia lacks rights to erasure or meaningful limits on data brokers, and transparency around algorithmic profiling remains minimal [11, 12]. This means that Australians inhabit a digital environment where trust in regulation is high but actual protections are partial.

Little empirical work has directly measured how Australians understand their rights, their risks, and their role in the data economy. Whereas European discourse is saturated by GDPR and U.S. discourse is shaped by skepticism of corporate surveillance, Australian privacy debates remain fragmented, reactive, and often shaped and dominated by legal professionals, policymakers, and technical experts rather than the broader public. Without systematic evidence of public understanding, policy reforms risk reinforcing existing misconceptions rather than addressing them.

This work addresses that gap through a large-scale survey of privacy knowledge, attitudes, and perceptions across Australia. To our knowledge, this is the first study to examine Australians’ understanding of digital privacy across both legal and technical dimensions, while also anchoring the findings in comparative data from the United States. Our study is guided by three research questions:

- (1) **RQ1:** How do Australians perceive the collection, sharing, and protection of their personal data online, and to what extent do they experience gaps between what they suspect and what they concretely understand?
- (2) **RQ2:** How do demographic factors (age, gender, education, income) shape not only privacy knowledge but also feelings of distrust, resignation, or agency in navigating digital privacy?
- (3) **RQ3:** How does Australians' fractured awareness of privacy protections compare with findings from the United States, and what do these contrasts reveal about designing privacy-respecting systems that bridge the gap between law, technology, and lived experience?

Answering these questions goes beyond mapping misconceptions. It reveals how systemic misunderstandings shape trust, agency, and equity in digital societies. Our findings provide an empirical foundation for reimagining privacy architectures, consent interfaces, and legal-technical alignments that move beyond compliance toward intelligibility. This highlights privacy not only as a technical software safeguard, but as a civic condition that determines whether digital infrastructures can maintain legitimacy, encourage trust, and empower citizens in an era of pervasive surveillance.

3 Methodology

This study employed a mixed-methods survey-based design to examine Australians' knowledge, perceptions, and misperceptions of digital privacy, focusing on awareness of national regulations, understanding of data flows, and attitudes toward personal information handling online. The goal was to produce robust, demographically representative, and internationally comparable results that can inform privacy-respecting software design and evidence-based policy reform. This mixed-methods approach integrates quantitative measures of knowledge and attitudes with qualitative narrative data, enabling both statistical analysis and thematic interpretation of participants' privacy perceptions. Our study received ethics approval from the authors' institutional review board (removed for double anonymous review).

3.1 Survey Design

The survey was deliberately designed to capture both factual knowledge and affective, interpretive, and socio-technical dimensions of digital privacy. Its design drew inspiration from Turow et al.'s large-scale U.S. study on Americans' misconceptions of data flows [42], though it was not a direct replication. Instead, we adopted a comparative framework that retained structural parallels to enable cross-national comparison, while incorporating new items tailored to the Australian regulatory landscape, sociocultural context, and technological ecosystem. In doing so, the instrument bridges international benchmarking with localized inquiry. Our final questionnaire consisted of 42 items, 41 structured questions and one open-ended narrative item organized into five thematic domains. Each domain combined knowledge testing, attitudinal measurement, and interpretive probes to capture the complexity of privacy understanding and sentiment:

Privacy Legislation Knowledge - Participants were asked to evaluate their familiarity with key Australian privacy laws and regulatory mechanisms (e.g., the Privacy Act 1988, the My Health

Record Act, the Consumer Data Right, and OAIC guidelines). Beyond binary true/false items, we included multiple-choice questions with plausible distractors to assess common misconceptions, and confidence ratings to capture whether correct answers reflected certainty or guesswork. This triangulation enabled us to distinguish genuine literacy from chance accuracy.

Online Tracking and Consent - These questions tested participants' understanding of technical tracking mechanisms (cookies, device fingerprinting, cross-site and cross-device profiling) and consent regimes (opt-in versus opt-out defaults, passive tracking, and dark patterns). Questions were embedded in everyday digital scenarios, such as online shopping, using streaming services, or connecting to public Wi-Fi, to explore how participants interpret and respond to real-world design cues. This experiential framing prioritizes lived dimensions over abstract technicalities.

Personal Data Perceptions - Participants were asked to identify the types of data they believe apps and platforms routinely access (location, health, financial, biometric, or communication metadata), and to reflect on how such data is stored, anonymized, sold, or shared with third parties. By combining factual and belief-oriented items, this domain highlights the fractured awareness that characterizes public perceptions, where individuals suspect that privacy violations occur, yet remain uncertain about their precise nature.

Trust, Concern and Agency - A series of Likert-scale questions probed how respondents perceive their control over personal data flows, their levels of concern or resignation, and their trust in various actors (government regulators, technology companies, health-care providers, and employers). These items sought to capture not just knowledge but also emotional orientations, such as distrust, apathy, or defiance, that shape everyday digital practices and acceptance of consent mechanisms.

Open-Ended Feedback - Finally, we invited participants to freely share experiences, stories, or metaphors that encapsulate their relationship to digital privacy, allowing us to capture rich accounts of frustration, anger, resignation, and personal narratives of disempowerment or resistance.

After every structured item, participants were also provided with a small optional text field where they could elaborate on their choice. Many respondents justified "Don't Know" selections by describing suspicion, distrust, or helplessness. These free-text explanations proved essential for demonstrating that high rates of uncertainty reflected fractured awareness rather than indifference, and they enriched the dataset with lived accounts that informed our thematic analysis in Section 4. To ensure accessibility across literacy levels, the survey instrument was designed in plain English, drawing on readability guidelines issued by the Australian Bureau of Statistics (ABS). We deliberately avoided legal or technical jargon, replacing terms like "cross-site tracking" with illustrative descriptions (e.g., "when companies follow what you do across different websites, even if you never log in"). Our goal was to elicit comprehension rather than test technical vocabulary.

To validate clarity and neutrality, the questionnaire underwent a two-stage pilot. In Stage one, 12 demographically diverse participants completed the survey while providing think-aloud commentary, which led to the simplification of several questions and the removal of ambiguities that could have biased responses. Stage two focused on tone and neutrality to avoid leading or emotive

language. The survey was intentionally designed to reflect privacy as a negotiation between law, technology, and lived experience, rather than a mere knowledge gap. The full questionnaire is available at https://docs.google.com/spreadsheets/d/1kbUCu4KeE_mCyyyVNjjtnK53Nvu6jMb_ for transparency and future reuse.

3.2 Recruitment Strategy

Recruitment was conducted entirely through organic, non-paid channels, reflecting a deliberate commitment to prioritize inclusivity, accessibility, and transparency, and avoiding reliance on commercial recruitment platforms. Two complementary recruitment strategies were employed: (i) **Social media outreach** - The survey link was shared across platforms such as Facebook, LinkedIn, and Twitter/X. Posts circulated organically through community networks and personal shares, without paid promotion or targeted advertising. This approach enabled the survey to reach a broad mix of participants, including individuals outside typical research participant pools. (ii) **Mailing lists** - The survey was also distributed via various research-related mailing lists. This method ensured engagement with participants who were already attuned to broader societal debates about technology and governance, while still allowing for diversity in digital knowledge and lived experience.

This recruitment strategy was designed to balance reach with representativeness, ensuring that the sample reflected a broad cross-section of Australian adults. While not a probability sample, the demographic composition aligns with national census patterns on gender, age, education, and income distributions, with slight over-sampling of younger and mid-income participants typical of online surveys. This balance supports cautious generalizability while still capturing the diversity of perspectives necessary for meaningful societal analysis. A total of 239 adult participants residing in Australia provided valid submissions to the survey. The final demographic breakdown is presented in Table 1.

Demographic Category	Distribution (%)
Gender	Man (43.5%), Woman (29.3%), Non-binary/Gender-diverse (4.6%), Prefer not to say (22.6%)
Age	18–30 (32.3%), 30–50 (29.7%), 50+ (21.3%), Prefer not to say (16.7%)
Education	Bachelor’s (31.4%), Postgraduate (15.9%), Voc/Some College (23.4%), High School or below (14.2%), Prefer not to say (15.1%)
Income	\$45k–120k (45.6%), Under \$45k (15.5%), \$120k–180k (12.1%), Over \$180k (4.6%), Prefer not to say (22.2%)

Table 1: Final Demographic Composition of Survey Cohort

3.3 Statistical Analysis

To examine demographic differences in privacy knowledge, perceptions, and attitudes, we conducted group-level comparisons across age, income, gender identity, and education. For categorical survey items, we report observed proportions (percentages) along with group sample sizes (n). Differences between groups were tested using Chi-squared tests of independence (χ^2), a non-parametric test appropriate for categorical data. The Chi-squared statistic quantifies how much the observed distribution of responses deviates from what would be expected if group membership had no effect. We

report the χ^2 value alongside a p -value, which indicates the probability that the observed differences could have arisen by chance. Following conventional thresholds, $p < .05$ is interpreted as statistically significant, $p < .01$ as highly significant, and $p < .001$ as very highly significant.

3.4 Comparative Anchoring with U.S. Study

To enable meaningful international comparison, the survey instrument was designed with comparative anchoring. Specifically, ten core questions were harmonized with Turow et al.’s 2023 U.S. study [42], focusing on foundational misconceptions about privacy policies, consent, and online tracking. Public misunderstanding in these areas affects both system design and regulatory effectiveness.

Local adaptation was carefully performed to preserve semantic equivalence while ensuring contextual relevance. For example: (i) U.S.-specific laws (e.g., HIPAA, COPPA) were replaced with Australian counterparts (e.g., the My Health Record Act, the Privacy Act 1988); (ii) References to “federal regulators” were reframed as “the Office of the Australian Information Commissioner (OAIC)” to enhance contextual clarity for local respondents; and (iii) Idiomatic phrases in U.S. survey items were rewritten in plain English, following ABS readability guidelines, to improve accessibility across literacy levels while maintaining semantic equivalence.

This comparative anchoring served two purposes. First, it situated Australian findings within a broader global discourse on digital privacy, allowing us to identify which misconceptions are globally persistent and which are shaped by national regulatory and cultural contexts. Second, it provided a baseline for assessing whether Australia’s comparatively privacy-aware regulatory environment has led to greater public understanding than in the U.S. As discussed in Section 4, our findings suggest the opposite. Australians exhibited similar levels of functional misunderstanding, while expressing higher, but often misplaced, trust in government regulation.

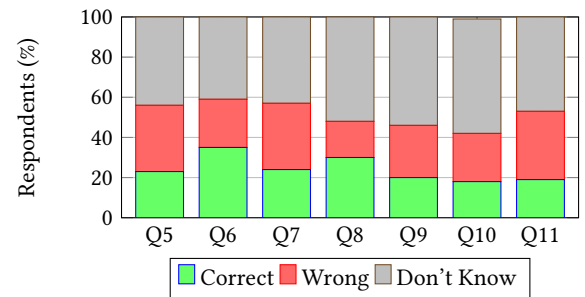


Figure 1: Knowledge questions (Q5–Q11): “Don’t Know” dominates across tracking and legal issues.

4 Results

Our analysis of survey responses from 239 Australian participants reveals a complex, multi-layered picture of public understanding of digital privacy. Rather than a simple knowledge deficit, the findings suggest a fractured landscape of suspicion, misplaced trust, everyday encounters with surveillance, and deeply felt emotions of resignation and distrust. In this section, we present the results across six interrelated themes that emerged from both quantitative survey items and qualitative open-ended feedback.

4.1 Fractured Awareness: Knowing Something is Wrong Without Knowing Why

One of the most consistent patterns in the data is the high prevalence of “Don’t Know” responses to core questions about online tracking and consent. For instance, 52.0% of participants selected “Don’t Know” when asked about cross-device profiling without logging (Q8), while 54.3% were uncertain whether privacy policies primarily protect companies or users (Q9). Similar uncertainty appeared in responses about spy pixels (Q7, 43.1%), Smart TV tracking (Q6, 41.5%), and breach notification laws (Q11, 46.7%). As shown in Figure 1, participants were more likely to answer “Don’t Know” than either “Yes” or “No” across all knowledge questions.

This pattern reflects what we term **fractured awareness**, a state in which individuals sense that something problematic may be occurring but lack the clarity to articulate or explain it. Rather than simple ignorance, it reflects a deeper cognitive and emotional dissonance: participants feel vulnerable and distrustful, yet cannot pinpoint the mechanisms of data collection, sharing, or surveillance.

⇒ **Insight.** High rates of “Don’t Know” responses reveal fractured awareness where users suspect problems but cannot articulate how surveillance operates, leaving them anxious yet disempowered.

4.2 The Illusion of Protection: Misplaced Trust in Privacy Policies and Regulation

Another major finding concerns the widespread misplaced trust in protective mechanisms. Only 19.9% of participants correctly identified that privacy policies primarily protect companies from liability (Q9), while a quarter mistakenly believed these documents shield users, and over half were unsure. Similarly, when asked whether Facebook privacy settings prevent advertisers from profiling (Q10), 57.2% responded “Don’t Know,” 24.4% incorrectly assumed protection, and only 18.4% correctly identified the limits of these settings. Even for questions tied to legal requirements, such as whether explicit opt-in consent is always required before tracking (Q18), less than one in five answered correctly.

These results reveal a pervasive “illusion of protection” where participants take comfort in the symbolic presence of privacy policies, consent toggles, warnings, and privacy laws, while overestimating their actual protective power. This illusion was echoed in open-ended comments like, “*I assumed the government wouldn’t let companies get away with this*” or “*If there are privacy settings, they must mean something*”. The design of user interfaces and the existence of legal frameworks encourage a false sense of security, sustaining software privacy engineering practices that are neither transparent nor effectively constrained.

⇒ **Insight.** Privacy policies and legal symbols create an illusion of protection, encouraging misplaced trust that masks the absence of real safeguards.

4.3 Emotional Landscape: Anxiety, Distrust, and Resignation

Open-ended responses revealed the emotional weight of privacy confusion. Participants used language of *anxiety* (“living in a glass

house”), *anger* (“they just do whatever they want”), and *resignation* (“nothing I can do, so why bother”). Q41 quantified this anxiety starkly: 68.4% of respondents rated their concern as extremely high (8, 9, or 10 on a 10-point scale), while only 13.0% reported being completely unconcerned (1 out of 10). As shown in Figure 2, the distribution skews heavily toward the upper end, indicating that privacy anxiety is not a fringe sentiment but a majority experience. These emotional responses are not peripheral; they shape how individuals engage with systems, policies, and consent mechanisms.

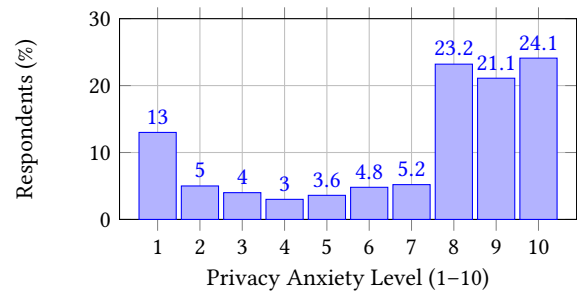


Figure 2: Privacy anxiety is widespread: 68.4% report extreme concern (8–10).

The qualitative responses to the open-ended prompt revealed diverse emotional reactions. Excerpts include:

- (1) “*I have stopped reading privacy policies. I know they are written to confuse me.*” (Age 34, Woman)
- (2) “*We have no control anymore. It is surveillance capitalism disguised as convenience.*” (Age 41, Non-binary)
- (3) “*Just tell me clearly what you’re doing with my info. Don’t bury it in legal jargon.*” (Age 51, Man)
- (4) “*I use incognito, VPNs and fake emails. It is a war out there.*” (Age 29, Woman)
- (5) “*Why doesn’t the government make this illegal?*” (Age 50, Prefer not to say)

The responses reflect a mix of resignation, anger, confusion, and pockets of digital resilience. They highlight the emotional toll of navigating opaque systems and reinforce the need for user-centered privacy design that prioritizes clarity, intelligibility, and trust.

⇒ **Insight.** Anxiety, distrust, and resignation are not side effects but measurable system outcomes that must be addressed in design and evaluation.

4.4 Everyday Surveillance in Plain Sight: Tracking, Devices, and Data Flows

Participants’ experiences with everyday technologies highlight the normalization of surveillance. Nearly half (44.4%) agreed that apps secretly listen through microphones (Q21), while another 45.0% were unsure. Similar patterns of uncertainty and suspicion emerged around Smart TV tracking (Q6), spy pixels in emails (Q7), and Wi-Fi surveillance (Q35). As summarized in Figure 3, “Don’t Know” responses consistently outnumbered correct identifications.

These findings suggest that participants perceive the effects of surveillance, such as eerily timed ads or uncanny personalization,

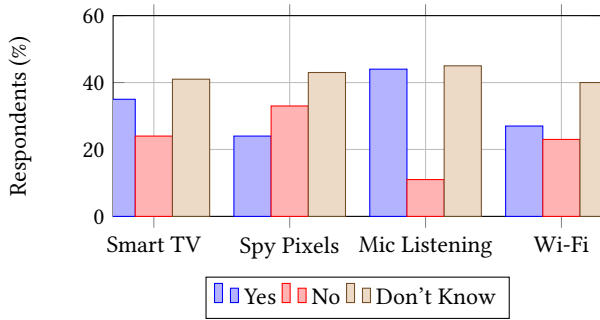


Figure 3: Everyday surveillance beliefs: most suspect hidden tracking, many remain unsure.

but struggle to explain the mechanisms behind them. As one respondent noted: “*I don’t know how, but I know they are listening*”. Such comments point to an experiential awareness where surveillance is “in plain sight”, visible through its effects but opaque in operation. This duality fosters both paranoia and resignation, as individuals come to expect tracking as a routine part of device use while feeling powerless to intervene.

⇒ **Insight.** Everyday surveillance is visible through its effects but opaque in operation, normalizing paranoia and resignation while eroding user agency.

4.5 Resignation Without Relief: Dependency, Distrust and the Limits of Agency

Questions probing attitudes toward agency and control revealed high levels of resignation. Nearly half of respondents (47.7%) strongly agreed that they worry about following the wrong steps to protect their data (Q25), while 39.4% strongly agreed that they had given up on controlling what companies learn about them (Q26). Similarly, 41.2% strongly agreed that companies invent new tracking methods faster than users can adapt (Q27). These findings echo Draper’s notion of *digital resignation* [14], but extend it by showing that resignation stems not only from fatigue but from fractured awareness.

⇒ **Insight.** Resignation emerges from fractured awareness and dependency without trust, revealing that privacy failures are as much affective as they are technical.

Trust metrics further underscore this fragility. Only 31.8% strongly agreed that they trust websites to guard their data as carefully as they would themselves (Q31), while 42.5% strongly agreed they trust themselves more than companies (Q28). Yet paradoxically, many respondents still rely on these companies for essential services. This reflects a condition of dependency without trust, a structural tension that reinforces resignation. These findings highlight that privacy failures are not only technical but also affective.

4.6 Stratified Misconceptions: Demographic Divides in Privacy Knowledge and Perception

Analysis by demographic groups revealed stark disparities. Older adults (50+, $n = 51$) were significantly more likely to select “Don’t

Know” across technical questions, averaging 58.2% uncertainty compared to 37.6% among younger participants aged 18–30 ($n = 77$, $\chi^2 = 14.7$, $p < .001$). In contrast, younger participants expressed higher suspicion of microphone listening (Q21: 54.1% of 18–30s vs. 29.4% of 50+, $\chi^2 = 9.8$, $p < .01$) and device tracking (Q8: 68.0% vs. 41.2%, $p < .01$). Lower-income respondents (under \$45k, $n = 37$) disproportionately reported resignation, with 62.5% agreeing they had “mostly or completely given up” on controlling data (Q26), compared to 39.1% in higher income brackets ($\chi^2 = 7.3$, $p < .05$). Non-binary participants ($n = 11$) reported elevated distrust in both companies and government, with 72.7% stating they “rarely or never” trust companies (Q33), compared to 61.4% overall. Similarly, respondents without tertiary education ($n = 34$) selected “Don’t Know” at rates above 60% for device fingerprinting (Q8) and health app data practices (Q20). These divides demonstrate that privacy misconceptions are not evenly distributed but intersect with age, education, and socioeconomic status. This stratification means that privacy harms are likely to fall disproportionately on those least equipped to navigate opaque systems, a finding that resonates with prior critiques of digital inequality.

⇒ **Insight.** Privacy misconceptions are stratified across age, income, education, and gender identity, placing greater burdens on the most vulnerable groups.

4.7 International Comparison: Contrasting Australian and U.S. Privacy Myths

By anchoring ten core questions to Turow et al. [42], we identified both shared misconceptions and distinct national differences. Like Americans, Australians demonstrated limited understanding of tracking mechanisms and placed inflated trust in privacy policies. However, Australians expressed significantly stronger belief in government protection, with comments such as “*Surely the laws cover that*”. This institutional trust differentiates the Australian context from the U.S., where skepticism toward both government and corporate surveillance is more pronounced.

Despite these differences, both populations exhibited fractured awareness and digital resignation, suggesting these are transnational phenomena rooted in the design of digital infrastructures rather than in cultural specifics. The comparative lens highlights the need for privacy engineering approaches that are both context-sensitive and globally relevant. Figure 4 illustrates these contrasts quantitatively, showing that Australians hold stronger beliefs in government protection, while both groups remain equally confused about cross-device tracking and the function of privacy policies.

⇒ **Insight.** Privacy engineering cannot be one-size-fits-all: what reassures users in one jurisdiction may deepen mistrust in another, making cultural sensitivity a design requirement.

4.8 Privacy Personas as Design Archetypes: The Misinformed, The Resigned and The Resistant

Our survey revealed three orientations toward privacy: the *Misinformed Majority*, who display misplaced trust and limited knowledge; the *Resigned Skeptics*, who recognize surveillance but feel

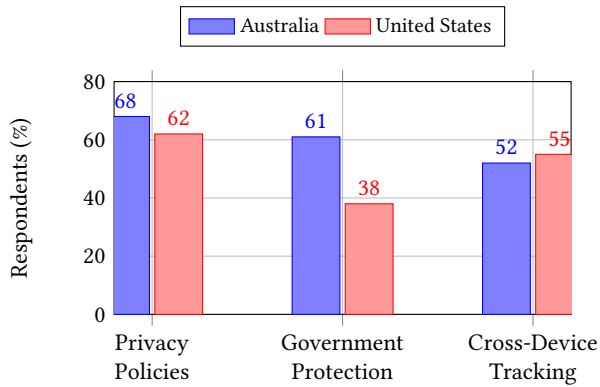


Figure 4: Australians show higher trust in government protection, while both groups remain confused about privacy policies and tracking.

powerless; and the *Aware Activists*, a minority who resist through VPNs, disposable emails, or other tools. These personas are not merely descriptive labels but design archetypes for socio-technical privacy engineering. For practitioners, they highlight divergent needs: the *Misinformed Majority* require clarity and simplification; the *Resigned Skeptics* require agency-restoring features such as refusal options and dashboards; and the *Aware Activists* require advanced controls without penalty. For researchers, the personas provide a framework to study how orientations evolve: how trust decays into resignation, and resignation sometimes transforms into resistance. Figure 5 visualizes these personas as a continuum, illustrating how misplaced trust often shifts into resignation, and in some cases, resistance.

⇒ **Insight.** Privacy personas should guide system design, ensuring tools adapt to the misinformed, restore agency for the resigned and empower activists without punishment.

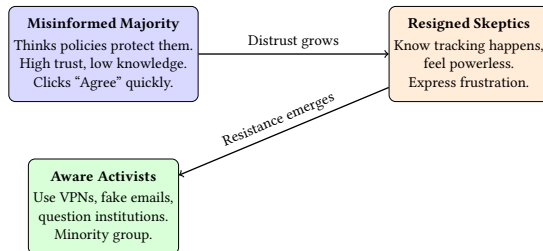


Figure 5: Privacy personas: Misinformed Majority, Resigned Skeptics, and Aware Activists.

5 Implications for Research and Practice

Our study reveals that Australians are not passive or indifferent to digital privacy. Instead, many inhabit a state of fractured awareness, sensing violations, distrusting platforms, and feeling manipulated, yet struggling to articulate or resist specific practices. This condition is more alarming than ignorance. It suggests that the systems we engineer do not merely fail to inform; they actively distort public understanding, producing citizens who are simultaneously alert and powerless.

For Software Engineering in Society (SEIS), this constitutes a crisis of civic agency. Building on the empirical findings in Section 4, the following implications offer actionable guidance for researchers, practitioners, and other stakeholders. Each theme, from fractured awareness to resignation, is reconceptualized as a challenge of design and governance that demands systemic intervention. These implications are grounded in observed patterns of misunderstanding, emotional response, and behavior across diverse demographic groups, rather than normative assumptions about ideal users.

5.1 Implications for Practice

5.1.1 Reframing Consent to Eliminate Dark-Pattern Engineering. Our findings demonstrate that most Australians perceive consent as manipulative. Over half (51.0%) could not correctly answer whether opt-in consent is legally required, and 57.2% doubted Facebook's settings had any real effect. Many participants admitted they clicked "accept all" not out of agreement but because refusal meant losing access. One respondent wrote, "It doesn't matter what button you click, they will do it anyway". These results expose consent interfaces as theatrical performances, simulating choice while funneling users toward compliance. This reveals a systemic problem where consent is reduced from a meaningful safeguard into a dark-pattern compliance ritual.

✓ **Recommendation.** Consent must be redesigned as a **communicative process rather than a checkbox for legal compliance**. For software engineers, this means *integrating intelligibility testing into requirements and QA pipelines*: does the interface communicate genuine alternatives, or does it merely simulate them? Designers should *employ layered explanations, refusal pathways that do not punish users, and visual feedback* that clarifies consequences of each choice. Regulators could mandate independent audits of consent interfaces to detect dark patterns, ensuring that the "choice" offered is substantive rather than symbolic. By reframing consent as a trust mechanism rather than a legal artefact, practitioners can restore agency and reduce resignation.

5.1.2 Designing with Personas in Mind. Our findings identified three privacy personas, the *Misinformed Majority*, the *Resigned Skeptics*, and the *Aware Activists*. Each group embodies different misconceptions, levels of trust, and forms of resistance, meaning that a single approach to privacy design will inevitably fail. For regulators, this means ensuring protections are understandable even to the *Misinformed Majority*, who otherwise over-trust laws and policies. For technologists, embedding comprehension checks must account for the *Resigned Skeptics*, who disengage unless refusal pathways are clear and empowering. For designers, advanced settings must be transparent and penalty-free, supporting *Aware Activists* who actively defend their privacy.

✓ **Recommendation:** Regulators, technologists, and designers should use privacy personas as design archetypes, tailoring protections to clarify for the misinformed, empower the resigned, and support activists without penalty.

5.1.3 Translating Policy into Legible Software Artefacts. When asked whether privacy policies protect companies or users, 54.3% said

they “weren’t sure” and 25.8% admitted complete ignorance. Nearly half believed hospitals must disclose breaches nationally, when in reality disclosure rules vary by jurisdiction. One participant reflected, “I thought the Privacy Act meant they can’t sell my data, clearly I was wrong”. These results show that the very mechanisms meant to reassure citizens, policies and laws, instead cultivate illusions of safety. Compliance alone is insufficient: a policy no one can understand is equivalent to no policy at all.

✓ **Recommendation. Policies and laws must be rendered legible through software artefacts that translate abstract protections into concrete user experiences of safety.** For software engineers, *this requires building interaction patterns that expose rights at the point of use*. For example, breach notifications embedded directly into healthcare apps rather than hidden in regulator websites. Designers should employ *plain-language dashboards, layered explanations, and jurisdiction-aware prompts* that reduce uncertainty. Regulators could mandate *intelligibility audits of policy-driven interfaces*, ensuring that compliance is measured not only by presence of a legal text but also by user comprehension. By embedding legibility into design and engineering processes, practitioners can transform laws from symbolic artefacts into lived safeguards.

5.1.4 Designing Against Resignation to Restore User Agency. Participants’ narratives of privacy as a “rigged game” or “a sinking boat with too many holes” highlight a dangerous affective outcome: resignation. Citizens do not simply mistrust platforms, they give up on protecting themselves. Quantitatively, 39.4% admitted they had “mostly given up” and 38.4% said they had “completely given up” on controlling what companies know about them. This is alarming. Systems that repeatedly frustrate or deceive do not merely fail, they condition disengagement, eroding agency and trust.

✓ **Recommendation. Interfaces must be engineered to counter resignation by embedding refusal pathways that do not punish users and by providing incremental empowerment tools rather than binary “all or nothing” choices.** For software engineers, this means *designing dashboards that visualize consequences of decisions* in real time, giving users a sense of tangible control. Designers can *integrate progressive disclosure*, showing small but meaningful steps to protect data, rather than overwhelming users with abstract settings. Regulators could require that *opt-out mechanisms be tested for usability* as rigorously as opt-in consent flows. By treating resignation as a system-level failure, engineers can create infrastructures that sustain engagement and rebuild user trust.

5.1.5 Advancing Equity by Designing for the Digitally Excluded. Our demographic analysis revealed striking inequities. Participants without tertiary education answered “Don’t Know” at rates above 60% for device fingerprinting and health app data practices. Older adults expressed helplessness: “I try to protect myself but I don’t know if I’m doing it right”. Lower-income respondents feared that trading privacy for discounts was unavoidable: “If it saves me money, I have no choice”. These disparities show that privacy confusion amplifies existing social divides, placing disproportionate burdens on those least equipped to navigate opaque systems.

✓ **Recommendation. Equity must be treated as a first-class design requirement, embedding multi-modal explanations, plain language, and culturally localized privacy tools into every system.** For software engineers, this means *adopting inclusive defaults*, such as voice-assisted explanations for older adults or icon-based cues for low-literacy groups. Designers can *apply participatory methods to co-create privacy features* with marginalized communities, ensuring tools resonate with diverse lived realities. Regulators could *mandate accessibility audits* for privacy controls, just as they do for physical accessibility in public infrastructure. Embedding equity at the core of privacy engineering ensures protections work for all users, not just the digitally fluent.

5.2 Implications for Research

5.2.1 Reframing Fractured Awareness to Address Engineered Confusion. Our data reveals that Australians are not uninformed but fractured in their awareness. They suspect wrongdoing yet cannot describe mechanisms such as fingerprinting or cross-device tracking. One participant noted: “I’m sure they do it, I just don’t know how”. This liminal state, knowing something is wrong but not being able to act, is qualitatively different from ignorance. Rather than a traditional “knowledge deficit”, fractured awareness reflects a condition of engineered confusion, where opaque systems actively obscure legibility and disable user agency.

✓ **Recommendation: Future research should operationalize fractured awareness as a measurable construct, developing metrics, experimental probes, and diagnostics to evaluate whether systems reduce, reinforce, or exploit confusion.** For SE, *fractured awareness should be treated as a system-level failure*, akin to accessibility or security flaws. Researchers could design experiments to test whether new consent mechanisms lower “Don’t Know” rates, or build diagnostic tools to benchmark the clarity of data-flow explanations. *Embedding fractured awareness into requirements engineering* would ensure that user intelligibility is systematically addressed throughout design, testing, and deployment.

5.2.2 Challenging Myths to Reduce Over-Trust in Software Systems. Australians displayed a distinctive myth: a belief that government regulation will protect them from privacy harms. Unlike Americans, who express skepticism toward both state and corporations, Australians over-trust institutional safeguards. One participant remarked, “I trust the government to stop the worst of it”, even as they admitted having “no idea what the rules are”. This misplaced trust creates systemic vulnerabilities, as users falsely assume protections exist and disengage from protective practices.

✓ **Recommendation: Researchers should treat myths as cultural infrastructure in privacy engineering, developing methods to surface, challenge, and design against misplaced trust.** For SE, this means *integrating cultural analysis into requirements elicitation and system evaluation*. Comparative studies can reveal whether myths of state protection are local anomalies or globally persistent illusions.

For example, a hospital consent app in Australia may require explicit prompts clarifying the legal limits of privacy protection, while in the U.S. interface design may place greater emphasis on explaining corporate data practices. *Embedding myth-awareness into design guidelines* ensures that privacy interfaces actively correct context-specific misconceptions, rather than unintentionally reinforcing them. This shifts privacy design from passive disclosure toward intentional clarification of user understanding.

5.2.3 Embedding Legibility to Make Law Executable in User Interfaces. The survey reveals a paradox, citizens believe laws protect them, even when they cannot name a single protection. Nearly half (46.7%) could not say whether hospitals are legally required to disclose breaches across states. One participant summarized the problem: “The rules are written for lawyers, not for people like me”. This disconnect between written law and lived understanding is not merely a legal issue; it is a SE challenge, since systems are the primary mediators of how rights are experienced in practice.

✓ **Recommendation: Legibility must be elevated to a first-class engineering requirement, with systems designed to render laws visible and comprehensible at the point of interaction.** For researchers, this calls for *methods to evaluate whether users not only comply but also understand their rights* when interacting with privacy features. For practitioners, this means *embedding interaction patterns, verification tools, and intelligibility tests* into requirements and QA pipelines. For example, a hospital consent app could provide layered explanations that explicitly clarify what regulations do and do not guarantee, with comprehension checks before consent is finalized. Such approaches move beyond legal compliance toward executable transparency, making law actionable for everyday users.

5.2.4 Integrating Emotional Impact into Software Engineering Outcomes. Anger, distrust, and resignation were consistent across our survey. “It’s impossible to win”, one participant wrote, while another admitted, “I’ve stopped caring what they know”. These affective outcomes are not merely peripheral side effects; they are measurable consequences of system design. Current evaluation practices in SE emphasize functionality, performance, and usability, but neglect the emotional sustainability of interactions. Our findings show that systems can actively generate feelings of helplessness and disengagement when they obscure, deceive, or overload users.

✓ **Recommendation: SE must integrate emotional impact assessments into requirements, design, and testing, ensuring that systems foster trust rather than resignation.** For researchers, this means *developing new evaluation frameworks that capture affective outcomes* through surveys, longitudinal studies, and controlled experiments. For practitioners, this *requires extending QA pipelines to include emotional impact checks* alongside usability testing. For example, a redesigned privacy dashboard could be tested not only for comprehension but also for whether users leave the interaction feeling empowered rather than defeated. By embedding emotional resilience as a design criterion, SE can prevent the erosion of trust and agency in digital infrastructures.

6 Threats to Validity

Internal Validity: While we sought demographic diversity through stratified quotas, participants ultimately self-selected into the survey, introducing bias. Those with stronger digital access or pre-existing privacy concerns were more likely to participate, potentially skewing results toward more privacy-aware individuals. Conversely, those most vulnerable to exploitation (e.g., with very low literacy or limited internet use) may be underrepresented. The online format also limits control over the respondent environment. Distractions, multitasking, or survey fatigue could have affected accuracy, particularly for more technical or legally nuanced items. Although we incorporated attention checks, and removed incomplete responses, satisficing behavior cannot be completely ruled out. Participants may unintentionally misstate their knowledge, overestimate protective behaviors, or underreport risky practices due to social desirability pressures. This is particularly relevant for questions about trust in government and institutions.

External Validity: Although our final cohort aligned with census-informed demographic distributions, inherent limitations of internet-based recruitment remain. Individuals without reliable connectivity, limited digital literacy, or from highly marginalized communities were less likely to participate. Yet these groups often face the greatest risks in opaque digital ecosystems, meaning our findings may underestimate the severity of misunderstandings at the societal margins. The survey was administered in English only, which may have excluded non-native speakers or individuals from culturally and linguistically diverse (CALD) or Indigenous backgrounds. These populations may conceptualize privacy differently, drawing on distinct cultural logics that our instrument could not capture. Future work should explicitly design localized instruments for multilingual and Indigenous contexts. Temporal dynamics also constrain generalizability as privacy awareness is shaped by fast-moving events, data breaches, high-profile scandals, or legal reforms. Finally, our comparative analysis with the United States must be interpreted cautiously. While we harmonized question wording with Turow et al.’s 2023 study to enable international comparison, cultural, legal, and linguistic differences complicate direct equivalence.

Construct Validity: Operationalizing privacy through survey instruments is inherently challenging, given its contested and evolving nature. While we used scenario-based true/false and multiple-choice items anchored in real-world practices (e.g., smart TVs, health apps, email tracking), simplification was unavoidable. Some items may have permitted multiple plausible interpretations, leading to inflated “Don’t Know” responses or misclassification of partial knowledge. Likert-scale questions captured concern, trust, and perceived agency, yet these constructs are context-sensitive and influenced by momentary emotion, recent media exposure, or personal experience. For instance, recent encounters with targeted advertising may heighten concern. Thus, our results indicate patterns rather than definitive measures of individual attitudes. Finally, while harmonization with U.S. instruments strengthens cross-national comparison, construct equivalence cannot be assumed. Terms such as “consent”, “tracking” or “breach notification” carry different connotations across jurisdictions. Future research should validate localized instruments that account for these semantic and cultural nuances.

7 Related Work

A substantial body of research has demonstrated that the public has limited knowledge of how personal data is collected, used, and shared. McDonald and Cranor’s seminal work [25] showed that reading all privacy policies a user encounters annually would require over 200 hours, highlighting the impossibility of informed consent under current practices. Solove’s taxonomy [38] further emphasized that privacy harms are often diffuse, cumulative, and intangible, making them difficult for users to recognize or resist. The so-called *privacy paradox*, which refers to the phenomenon where users express concern about privacy yet readily disclose personal information, has been widely studied [2, 5, 30]. However, recent work has critiqued the paradox framing for oversimplifying user behavior and ignoring structural constraints [14, 40]. Our study advances this critique by showing that behavioral dissonance often stems from *misconceptions* about protections that do not exist. Rather than indifference, Australians in our sample exhibited what we term *fractured awareness*: knowing something is wrong but lacking the conceptual tools to articulate why.

Cross-national differences in privacy perceptions have long been theorized [7, 20], but empirical user-level studies remain scarce. Turow et al. [42] surveyed over 2,000 U.S. participants and found pervasive misconceptions about HIPAA, COPPA, and other laws. Our study adapts their instrument to the Australian context, exposing parallel misunderstandings: Australians similarly overestimate their protections but differ in the locus of their trust, placing more faith in government regulation compared to Americans’ greater skepticism. This aligns with comparative privacy scholarship that emphasizes national *privacy cultures*, the shared expectations shaped by regulatory history, institutional trust, and public discourse [7, 19]. By providing harmonized data, our study not only enables cross-country comparison but also offers insight into how legal context and cultural narratives produce distinct forms of misconception.

A growing literature critiques how consent is operationalized in digital interfaces. Gray et al. [18] documented *dark patterns* that exploit cognitive biases to nudge users toward disclosure. Nouwens et al. [31] found that GDPR-compliant banners often systematically favor consent, undermining meaningful choice. Mathur et al. [26] further catalogued manipulative design tactics in consent flows, coining the term *consent theatre* [35] to describe the illusion of control without actual agency. Design frameworks exist: Schaub et al. [36] advocate layered, contextual, and actionable notices, while Utz et al. [44] demonstrated comprehension improvements via redesigned cookie banners, but implementation remains inconsistent.

The intersection of SE and ethics has received increasing attention, with recent scholarship stressing the civic responsibility of engineers to embed transparency, equity, and accountability into systems [4, 15]. Scholars argue that privacy-preserving technologies must be intelligible to end-users, not only technically robust [11, 37]. Our work contributes to this discourse by empirically documenting how legal complexity and poor design combine to create widespread misperception and fractured awareness. In contrast to studies that emphasize developer intentions or regulatory compliance, we highlight the *end-user knowledge gap* as a systemic failure of SE practice. Privacy interfaces, we argue, are not peripheral UX

features but central socio-technical infrastructures that mediate digital citizenship.

Prior work has largely focused on regulatory analysis, digital rights literacy, or experimental interface studies. Our contribution lies in combining a large-scale, demographically balanced survey that measures both technical and legal knowledge of privacy in Australia; open-ended qualitative responses that capture the emotional dimensions of privacy such as resignation, distrust, and suspicion often absent in quantitative studies; cross-national benchmarking that contrasts Australian misconceptions with U.S. findings, surfacing culturally distinct myths and misplaced trust patterns; and implications for software engineers framed through concepts like *fractured awareness*, *misplaced trust*, and *privacy-literacy-by-design*, which offer new theoretical and practical tools for rethinking system design. In doing so, our study extends prior research while reframing privacy not simply as a matter of regulation or HCI, but as a socio-technical challenge central to the legitimacy and equity of software-mediated societies. It also provides a methodological framework that can be replicated in other jurisdictions, enabling a more globally informed evidence base for privacy-aware engineering.

8 Summary

This paper presented the first nationally representative study of Australians’ privacy knowledge, attitudes, and misconceptions, focusing on how regulation, technical tracking, and consent design shape everyday digital life. Based on a demographically balanced survey of 239 participants, our mixed-methods analysis revealed the scale and emotional depth of privacy misunderstandings in an advanced democratic context. Our findings show a systemic privacy literacy gap. Most participants mistakenly believed that privacy policies prevent data sale, that the Privacy Act 1988 broadly protects against behavioral tracking, and that consent interfaces meaningfully limit surveillance. Knowledge of mechanisms such as device fingerprinting and opt-in consent was low, with “Don’t Know” responses often exceeding 40–50%. This reflects fractured awareness: people sense violations are occurring but cannot describe or resist them. Qualitative narratives reinforced this picture, depicting privacy as a “rigged game” that fuels resignation, frustration, and mistrust. Cross-national comparison with U.S. data revealed both common myths and distinctive Australian patterns, particularly higher trust in government regulation. Such misplaced confidence amplifies opacity and erodes agency. Overall, digital privacy is not as a legal or technical checkbox, but is a key socio-technical challenge rooted in comprehension, communication, and trust. Addressing fractured awareness requires rethinking privacy-by-design as privacy-literacy-by-design, embedding intelligibility, inclusivity, and cultural sensitivity into systems. For software engineers, policymakers, and designers, this means building privacy mechanisms as civic communication infrastructures that empower users, rather than reinforce myths or resignation.

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